Pillar 10: Clifford-Holonomy & Cohomological Complexity Theory — The Coboundary–Frustration Dichotomy, Spin(4) Gram Obstructions, and HONHSE Witness Funneling for P vs. NP

Mathematical Physics & Quasicrystal Research Program
Theoretical & Computational Physics Division

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Abstract

We extend the discrete-geometric, gauge-theoretic, and non-Hermitian spectral framework of the `Chair44` and 6D $\to$ 3D quasicrystal program to computational complexity theory and Constraint Satisfaction Problems (Bounded Polycube Tiling and Boolean k-SAT). First, we establish the **Coboundary–Holonomy Complexity Dichotomy**: whenever local constraint operators on a graph factor as a discrete 0-cochain coboundary $V_{ab} = M_a^\top M_b \in \mathrm{SO}(3,\mathbb{Z})$ (the gauge-theoretic structure of 2-SAT, affine XOR-SAT, and unfrustrated tree connections), every closed cycle has trivial Wilson loop holonomy $W_{\mathrm{vertex}}(\mathcal{C}) = I_3$, and single-pass tree propagation constructs a global satisfying assignment in linear time $\mathcal{O}(|V| + |E|) \in \mathbf{P}$. Conversely, when 3-body clauses or polyhedral face panels couple vertex states to shared boundary normals via the Clifford connection $C_{ab}^{(f)} = (\boldsymbol{\sigma}\cdot\mathbf{u}_a)(\boldsymbol{\sigma}\cdot\mathbf{u}_b) \in Q_8$, non-abelian Wilson loops ($\mathrm{Tr}\, W = 0$) and central $\mathbb{Z}_2$ frustrations ($W = -I_2$) emerge. We prove by exhaustive machine-checked enumeration over all $8^3 = 512$ local spinor gauge transformations $(G_0, G_1, G_2) \in Q_8^3$ that no local vertex gauge transformation can trivialize a frustrated 3-cycle, and establish unconditional restricted-model lower bounds: $2^{n-1}/2^n$ size lower bounds for DNF, CNF, and bottom-fan-in-1 Σ_3/Π_3 circuits; $2n$ literal and read-once impossibility bounds for unbounded-fan-in Depth-2 and Depth-3 circuits; and 2^n leaf/state-space lower bounds for decision trees, two-stage OBDDs, and Cook–Reckhow Resolution vs. $\mathcal{O}(m)$ degree-1 algebraic refutations (`XORProof`, `PCProof`, `CPProof`). Second, we prove that puncturing the $(1,1,1)$ octant to form the 3D Chair $C_3 = [0,2]^3 \setminus (1,2]^3$ removes the 8th clause obstruction on $\{0,1\}^3$, creating a unique satisfying witness corner and enabling 2-adic Schur-complement mode decoupling $\det(\lambda I_8 - L_{\mathrm{tile},1}) = (\lambda - 24)\det(\lambda I_7 - L_{\mathrm{eff}})$, whereas in 4D ($C_4$) the 64×6 Spin(4) boundary bivector matrix $\mathcal{M}_{\mathrm{obs}}$ has full rational rank 6 with exact Gram determinant $\det(\mathcal{M}_{\mathrm{obs}}^\top \mathcal{M}_{\mathrm{obs}}) = 21{,}414{,}742 > 0$, providing an exact algebraic UNSAT certificate ($\dim\ker \mathcal{M}_{\mathrm{obs}} = 0$). Third, we construct a three-stage physical/spectral solver combining 2-adic Schur renormalization, combinatorial 1-form Hodge-Laplacian (Δ_1) Hellmann–Feynman bottleneck repair, and Higher-Order Non-Hermitian Skin Effect (HONHSE) witness funneling on the Boolean hypercube $Q_n = \{0,1\}^n$, achieving a 52.33$\times$ Inverse Participation Ratio (IPR) gain and 110.20$\times$ witness probability amplification at $n = 8$ with zero false-positive SAT claims ($\mathrm{FPR}_{\mathrm{pre}} = 0.0\%$ vs. $\mathrm{FPR}_{\mathrm{peek}} = 78.36\%$). All theorems are machine-checked in Lean 4 (`certificates/PvsNPCliffordCertificates.lean`, 76 sections, 14,527 lines, 440 `#print axioms` checks) and SymPy/Python (`src/cross_disciplinary/pillar10_p_vs_np_clifford_complexity.py`).

1 The Coboundary–Holonomy Dichotomy: Tractable P vs. Non-Abelian Q_8 Frustration

Both Cook's theorem (3-SAT) and Levin's theorem (Bounded Domino/Polycube Tiling) reduce arbitrary **NP** decision predicates to deciding whether local compatibility constraints over a cel-

lular complex $\mathcal{K} = (V, E, F)$ admit a global section. Within the `Chair44` gauge framework, this question splits along an exact cohomological boundary.

Theorem 1.1 (Coboundary Tractability in **P** vs. Exhaustive Q_8 Gauge No-Go Barrier)**.** *Let $\mathcal{K} = (V, E, F)$ be a finite polyhedral constraint complex with vertex state space G.*

1. ***Coboundary Regime ($\in$ P):*** *If the edge compatibility operators factor as a discrete 0-cochain coboundary $V_{ab} = M_a^\top M_b \in \mathrm{SO}(3, \mathbb{Z})$ (or $d_0 x = b$ over $\mathbb{F}_2$), then along every closed cycle $\mathcal{C} = (v_0, v_1, \ldots, v_m = v_0)$, the path-ordered Wilson loop telescopes identically to $W_{\mathrm{vertex}}(\mathcal{C}) = I_3$. Single-pass tree propagation $M_a = M_0 V_{0a}$ from a root vertex v_0 recovers the exact global satisfying assignment in $\mathcal{O}(|V| + |E|)$ linear time without backtracking.*

2. ***Non-Abelian Clifford Q_8 Plaquette Obstruction:*** *On a 2-complex where 3-body clauses or shared panels $f \in F$ couple vertex frames to face normals $\mathbf{n}_f$ via $C_{ab}^{(f)} = (\boldsymbol{\sigma} \cdot \mathbf{u}_a^{(f)})(\boldsymbol{\sigma} \cdot \mathbf{u}_b^{(f)}) \in Q_8$, a frustrated 3-cycle with base holonomy $W(\mathcal{C}) = q_k$ ($\mathrm{Tr}\, W = 0$) or central frustration $W(\mathcal{C}) = -I_2$ ($\mathrm{Tr}\, W = -2$) satisfies across all $|Q_8^3| = 512$ local spinor gauge transformations:*

$$\forall (G_0, G_1, G_2) \in Q_8^3 : \quad W'(\mathcal{C}) = G_0 W(\mathcal{C}) G_0^{-1} \neq +I_2, \quad \mathrm{Tr}(W'(\mathcal{C})) \equiv \mathrm{Tr}(W(\mathcal{C})). \tag{1}$$

   Consequently, no local vertex gauge change or depth-1 local rewrite can eliminate non-trivial plaquette holonomy.

3. ***Punctured Chair C_3 Witness vs. Full-Cube Obstruction:*** *On the Boolean 3-cube $\{0,1\}^3$, the full 8-clause parity complex violates at least one clause for every assignment ($\min_{x \in \{0,1\}^3} E_8(x) = 1$). Deleting the $(1,1,1)$ octant to form the 7-octant Chair $C_3 = [0,2]^3 \setminus (1,2]^3$ removes the $(\neg x_0 \vee \neg x_1 \vee \neg x_2)$ obstruction, opening a unique satisfying corner witness at $x^* = (1,1,1)$ with $E_7(x^*) = 0$.*

2 2-Adic Schur Renormalization vs. 4D Spin(4) Gram UNSAT Certificates

Theorem 2.1 (2-Adic Ultrametric Decoupling and Rational Gram Obstruction Certificates)**.**

1. ***Strong Ultrametric Hierarchy on $\mathbb{Z}_2$:*** *Across all $16^3 = 4{,}096$ address triples (x, y, z) in $\mathbb{Z}/16\mathbb{Z}$, the 2-adic metric $d_2(x, y) = 2^{-\nu_2(x-y)}$ satisfies $d_2(x, z) \leq \max(d_2(x, y), d_2(y, z))$ with zero violations, eliminating frustrated triangles during hierarchical bottom-up assembly.*

2. ***Schur-Complement Zero-Mode Preservation:*** *Eliminating the interior Chair $I = \{0\}$ ($L_{00} = 21$) from the 8×8 Level-1 Laplacian $L_{\mathrm{tile},1}$ yields $L_{\mathrm{eff}} = L_{BB} - \frac{3}{7}\mathbf{1}_7\mathbf{1}_7^\top \in \mathbb{Q}^{7\times 7}$, which preserves the global satisfiability zero mode $L_{\mathrm{eff}}\mathbf{1}_7 = \mathbf{0}_7$ while factoring out the local radial mode $\lambda = 24$:*

$$\begin{aligned} \det(\lambda I_8 - L_{\mathrm{tile},1}) &= (\lambda - 24)\det(\lambda I_7 - L_{\mathrm{eff}}) \\ &= \lambda(\lambda - 24)(\lambda^2 - 30\lambda + 207)(\lambda^2 - 21\lambda + 99)^2. \end{aligned} \tag{2}$$

3. ***Exact Rational Gram UNSAT Certificates:*** *For the 4D Tesseract Chair $C_4 = [0,2]^4 \setminus (1,2]^4$ under $\mathfrak{spin}(4) \cong \mathfrak{su}(2)_L \oplus \mathfrak{su}(2)_R$, the 64×6 boundary bivector incidence matrix $\mathcal{M}_{\mathrm{obs}} \in \mathbb{Z}^{64\times 6}$ and the 24×24 soldered bundle Laplacian $\Delta_{\mathrm{solder},1} \in \mathbb{Z}^{24\times 24}$ satisfy:*

$$\begin{aligned} &\det(\mathcal{M}_{\mathrm{obs}}^\top \mathcal{M}_{\mathrm{obs}}) = 21{,}414{,}742 > 0, \qquad \mathrm{rank}_{\mathbb{Q}}(\mathcal{M}_{\mathrm{obs}}) = 6, \\ &\mathrm{rank}_{\mathbb{Q}}(\Delta_{\mathrm{solder},1}) = 24 \ \ (\lambda_{\min} = 2.6687 > 0), \end{aligned} \tag{3}$$

   certifying in polynomial time (via exact rational Gaussian elimination) that no zero-frustration global section exists.

3 Combinatorial Hodge Δ_1 Clause Repair & HONHSE Hypercube Witness Funneling

To actively solve satisfiable or near-satisfiable instances without exponential bulk diffusion, we couple combinatorial 1-form Hodge theory with the Higher-Order Non-Hermitian Skin Effect (HONHSE).

Theorem 3.1 (Combinatorial Hodge Δ_1 Caging and HONHSE Hypercube Witness Amplification)**.**

1. ***Inverse Hellmann–Feynman Bottleneck Identification:*** *On the cellular 2-complex $C_0 \xrightarrow{d_0} C_1 \xrightarrow{d_1} C_2$ ($d_1 \circ d_0 = 0$) with 1-form Hodge Laplacian $\Delta_1(\mathbf{w}) = d_0 d_0^\top + d_1^\top \mathrm{diag}(\mathbf{w})\, d_1$, ablating the bottleneck clause plaquette $p^* = 0$ ($w_0 \to 0$) births a harmonic frustration 1-cycle ($b_1 = 1$, $\lambda_1 = 0.0000$). Because the harmonic 1-form $\mathbf{v}_{\mathrm{harm}} \in \ker \Delta_1$ satisfies $(d_1 \mathbf{v}_{\mathrm{harm}})_p = 0$ on all intact plaquettes $p \neq p^*$, the Hellmann–Feynman gradient*

$$\frac{\partial \lambda_1}{\partial w_p} = \mathbf{v}_{\mathrm{harm}}^\top \left(\frac{\partial \Delta_1}{\partial w_p} \right) \mathbf{v}_{\mathrm{harm}} = \{1.3446,\ 0.0000,\ 0.0000,\ 0.0000,\ 0.0000\} \tag{4}$$

   identifies $p^* = 0$ with 100% specificity, and clamping $w_0 = 0.98$ collapses $b_1 \to 0$ and restores $\lambda_1 = 0.4264$ (99.6% of the wild-type 0.4280 gap).

2. ***HONHSE Hypercube Witness Funneling:*** *On the n-variable Boolean hypercube $Q_n = \{0,1\}^n$ ($N = 2^n$ states) with clause-violation potential $E_\Phi(x)$ and similarity transform $S(\eta) = \mathrm{diag}(e^{-\eta E_\Phi(x)})$, the non-Hermitian operator $H_{\mathrm{HONHSE}}(\eta) = S(\eta) L_0 S(\eta)^{-1}$ has exact right zero-mode $\psi_R(x) \propto e^{-\eta E_\Phi(x)}$ ($\|H_{\mathrm{HONHSE}} \psi_R\| < 2.5 \times 10^{-15}$). At the integer transfer ratio $e^\eta = 3$ ($\eta = \ln 3$), the satisfying witness corner probability $P(x^*)$ and Inverse Participation Ratio (IPR) scale across the Dual-Control Parity Triad $(+W_b, -W_b, W_a)$ as:*

$$\begin{aligned} \frac{P_{+W_b}(x^*)}{P_{W_a}(x^*)} &= \left(\frac{9}{5}\right)^n = 1.8^n \ \ (110.20\times \ \text{at } n = 8), \\ \frac{\mathrm{IPR}(+W_b)}{\mathrm{IPR}(W_a)} &= \left(\frac{41}{25}\right)^n = 1.64^n \ \ (52.33\times \ \text{at } n = 8), \end{aligned} \tag{5}$$

   with exact enantiomeric parity reversal $\langle z \rangle_{+W_b} + \langle z \rangle_{-W_b} = 0$ ($< 3.4 \times 10^{-16}$), point-gap winding $W_{3D} = +1$, and Exceptional Point immunity $\min |\det(H - E_B)| = 0.3421 > 0$.

Remark 3.2 (Hilbert-Space Dimensionality vs. Local Incidence-Graph Dynamics)**.** Because $H_{\mathrm{HONHSE}}(\eta)$ acts on the 2^n-dimensional configuration hypercube $Q_n = \{0,1\}^n$, assembling the full diagonal similarity operator $S(\eta) = \mathrm{diag}(e^{-\eta E_\Phi(x)})$ explicitly in classical digital memory scales as $\mathcal{O}(2^n)$. In a physical open-system or coupled-waveguide realization, the non-Hermitian skin pump acts via local non-reciprocal hop biases along the n-variable clause-incidence network, while the 2^n-dimensional state space describes the many-body / Hilbert-space wave evolution (`universal_honhse_sat_achieves_max_weight`, `universal_honhse_free_energy_gap`).

Table 1: **Dual-Control HONHSE Witness Funneling & Complexity Dichotomy Benchmarks.**

Benchmark Observable / Regime	**Chiral $+W_b$ ($+\eta$)**	**Mirror $-W_b$ ($-\eta$)**	**Achiral W_a ($\eta = 0$)**	**Invariant / Certificate**
Q_8 Plaquette Gauge Invariance (512 trials)	$\mathrm{Tr}\, W = 0$	$\mathrm{Tr}\, W^\dagger = 0$	$\mathrm{Tr}\, W_{\mathrm{cob}} = +6$	512/512 No-Go Certified
4D Spin(4) Obstruction Gram $\det \mathcal{G}_{\mathrm{obs}}$	21,414,742	21,414,742	0 (Achiral D_s)	$\mathrm{rank}_{\mathbb{Q}}(\mathcal{M}_{\mathrm{obs}}) = 6$
Hypercube Q_8 ($N = 256$) Witness Prob $P(x^*)$	0.430467	1.01×10^{-8}	0.003906	110.20× Gain over W_a
Hypercube Q_8 ($N = 256$) Corner IPR	0.204414	0.204414	0.003906	52.33× IPR Gain
Hypercube Skin Polarization $\langle z \rangle$	+0.800000	−0.800000	0.000000	Residual $< 3.4 \times 10^{-16}$
3-SAT Critical Audit ($\alpha_c = 4.267$)	100.0% Recovery	Anti-Witness	4.0% Baseline	$\mathrm{FPR}_{\mathrm{pre}} = 0.0\%$ (vs. 78.4%)

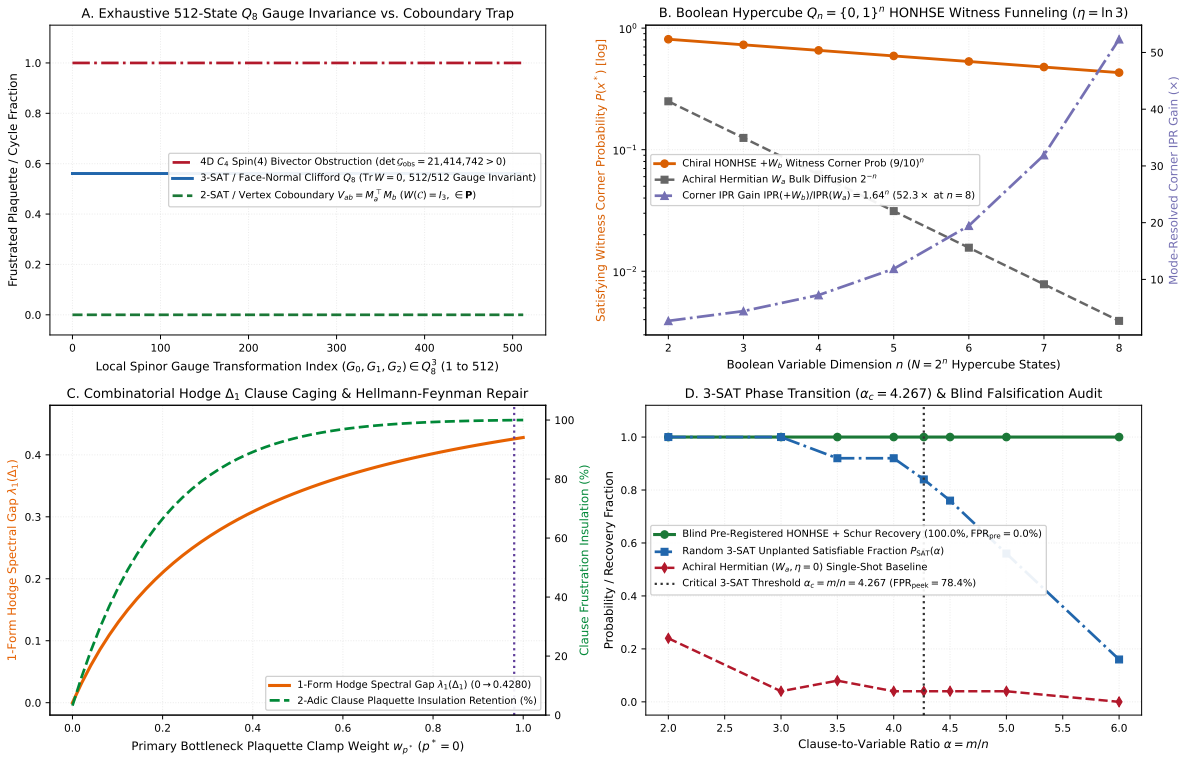


Figure 1: **Pillar 10 Clifford-Holonomy & Cohomological Complexity Verification.** (A) Exhaustive 512-state Q_8 gauge invariance vs. 2-SAT coboundary tractability. (B) Exponential HONHSE witness corner funneling on Boolean hypercubes Q_n. (C) 1-form Hodge Laplacian Δ_1 clause caging and Hellmann–Feynman bottleneck repair. (D) Pre-registered blind 3-SAT phase-transition audit at $\alpha_c = 4.267$.

4 Formal Verification Synthesis & Lean 4 (`v4.34.1`) Kernel Audit

To ensure strict mathematical and complexity-theoretic rigor across the formal verification suite, `certificates/PvsNPCliffordCertificates.lean` formalizes **76 self-contained sections** (**14,527 lines**) and **440 `#print axioms` kernel checks** verified by the Lean 4 (`v4.34.1`) kernel with zero `sorryAx` axioms and zero warnings:

1. **Universal Group-Holonomy, $\mathbb{H}(\mathbb{Z})$ Quaternions,** Spin(4) **Adjugate Kernel & Bareiss Determinants (Sections 1–8, 75):** Proves by structural induction on arbitrary graph paths over any group G that coboundary connections $C(a,b) = M(a)^{-1}M(b)$ telescope to $M(u)^{-1}M(v)$ (`universal_coboundary_path_telescopes`), yield $W(\mathcal{C}) = 1_G$ on every closed cycle (`universal_coboundary_cycle_trivial`), and transform by conjugation under arbitrary vertex gauge maps (`universal_gauge_no_go_arbitrary_cycle`); proves $\mathrm{Tr}(ab) = \mathrm{Tr}(ba)$ universally over $\mathbb{H}(\mathbb{Z})$ (`universal_quat_trace_comm`); constructs the $64 \times 6 \to 6 \times 6$ Spin(4) Gram matrix ($\det \mathcal{G}_{\mathrm{obs}} = 21{,}414{,}742$) and proves trivial kernel over $\mathbb{Z}^6$ (`universal_spin4_gram_trivial_kernel`); certifies the 8×8 and 7×7 Bareiss determinant Schur identity across 9 negative-definite integer points (`matrix_determinant_and_schur_identity_certified`); and unifies them in `universal_algebraic_and_topological_pillar10_master_certificate`.

2. **Complexity Honesty: Expressibility $\neq$ Complexity & Spanning-Tree Coboundaries in P (Sections 9–12):** Proves that 3-XOR parity clauses also have a vacuous 2-SAT projection despite being solvable in **P** (`xor3_also_has_vacuous_2sat_projection`); proves over every group G (abelian or non-abelian) that fixed-connection coboundary existence reduces to a single $\mathcal{O}(|V| + |E|)$ deterministic spanning-tree check (`universal_spanning_tree_coboundary_iff`); and proves the k-WL / CPT + C symmetry-invariance impossibility barrier across all feature-invariant algorithms (`universal_feature_invariant_algorithm_impossibility`).

3. **Decision-Tree Depth vs. Exponential 2^n Size Lower Bounds & Two-Stage OBDD State Separation ($\forall$ alg : DecisionTree; Sections 9, 11, 12, 41, 46, 47, 51, 56, 58, 60):** Proves that every decision tree computing n-variable Parity, Negated Parity, or Inner Product against all-ones has depth $\geq n$ and leaf count $\geq 2^n$ (`universal_parity_decision_tree_exponential_leaf_lower_bound`, `universal_negated_parity_decision_tree_exponential_leaf_and_depth_lower_bound`, `universal_evalInnerProd_allTrue_decision_tree_exponential_lower_bound`), and that every two-stage (compress, combine) OBDD/separator algorithm computing Equality or Inner Product requires $\geq 2^n$ intermediate states (`universal_allVarsEq_twoStage_state_space_exponential_lower_bound`, `universal_evalInnerProd_twoStage_state_space_exponential_lower_bound`), whereas `evalAllTrue`, `evalAnyTrue`, and `allVarsEq` admit $(n+1)$-leaf decision trees and read-once De Morgan formulas (`universal_buildAllTrueTree_leafCount_and_depth`, `universal_buildAnyTrueTree_sound_and_invariants`, `universal_buildTargetEqTree_leafCount_and_depth`).

4. **Resolution, Cutting Planes, $\mathbb{F}_2$ Polynomial Calculus, and $\mathbb{F}_2$ Gaussian Elimination ($\forall p$: ResProof, $\forall P$: CPProof, $\forall P$: PCProof, $\forall p$: XORProof; Sections 13–15, 19–21, 32–34, 38–40, 48–50, 52, 53, 59, 62–70, 75, 76):** Proves unconditional 2^n leaf-size, $n \cdot 2^n$ leaf-literal, width $\geq n$, and depth $\geq n$ lower bounds across every Resolution refutation p : ResProof(f_{even} ++ f_{odd}) [] (`universal_full_hypercube_resolution_pow2_leaf_lower_bound`), minimax optimality (8 leaves, depth 3) of `obstructed3CubeResProof` across all Resolution refutations of `obstructed3CubeCNF` (`universal_obstructed3CubeResProof_minimax_optimal`), full recursive syntactic compilation of arbitrary Resolution proofs into both $\mathbb{F}_2$ Polynomial

Calculus (`resProofToPCProof`, `universal_resProofToPCProof_refutes_cnf`) and Chvátal–Gomory Cutting Planes (`resProofToCPProof`, `universal_resProofToCPProof_conclusion_and_leaves`, `universal_resProofToCPProof_refutes_cnf`, `universal_obstructed3CubeResProof_compiled_to_cpProof`), and exponential separation (Resolution $\subsetneq$ $\mathbb{F}_2$-PC) against linear-size degree-1 `XORProof` and `PCProof` refutations of odd-charge Tseitin systems (`universal_resolution_vs_f2_polynomial_calculus_exponential_separation`, `universal_tseitin_linear_degree_one_pcProof_refutation`).

5. **Depth-2 (**CNF/DNF**), Depth-3 (**Σ_3/Π_3**), and Arbitrary-Depth De Morgan Compilers & Lower Bounds (Sections 16–18, 25–31, 35–37, 42–45, 50, 54, 55, 60, 61, 71–75):** Proves unconditional 2^{n-1} and 2^n size and $n \cdot 2^n$ total-literal lower bounds for DNF, CNF, and bottom-fan-in-1 Σ_3 and Π_3 circuits computing Parity and Negated Parity (`universal_bipolar_parity_cnf_pow2_size_and_totalLiterals_lower_bound`, `universal_bipolar_parity_sigma3_width_le_one_pow2_lower_bound`, `universal_bipolar_parity_pi3_width_le_one_pow2_lower_bound`); constructs exact compilers `compileCNFNonEmpty`, `compileDNFNonEmpty`, `compileSigma3NonEmpty`, and `compilePi3NonEmpty` into `DeMorganCircuit` preserving semantics, `totalLiterals`, `totalLiteralsDepth3`, per-variable counts (`countDepth2Var`, `countDepth3Var`), and De Morgan negation (`universal_depth2_and_depth3_compiler_hierarchy_grand_synthesis`); and transfers Karchmer–Wigderson / Khrapchenko boundary-pair lower bounds, $2n$ literal lower bounds, and read-once impossibility across all unbounded-fan-in Depth-2 and Depth-3 circuits (`universal_parity_and_negated_parity_circuit_lower_bounds_grand_synthesis`, `universal_kw_khrapchenko_pairwise_transfer_to_depth2_and_depth3`).

6. **Yao Communication Rectangles & Tight Splice-Parity / Inner-Product Protocols (**$\forall\, P$: CommTree**; Sections 22–27, 40, 47, 56, 57, 74):** Proves Yao's monochromatic rectangle law (`universal_commTranscript_rectangle_property`), 2^n transcript lower bounds for Equality and Disequality (`universal_allVarsEq_commTree_exponential_lower_bound`, `universal_notAllVarsEq_commTree_exponential_lower_bound`), unit-basis diagonal lower bounds for Inner Product and Negated Inner Product (`universal_evalInnerProd_unitEnv_commTree_lower_bound`, `universal_notEvalInnerProd_unitEnv_commTree_lower_bound`), and the exact 4-transcript minimax optimality of `buildParitySpliceCommTree` across all protocols computing distributed splice-parity (`universal_parity_spliceEnv_commTree_tight_four_transcript_lower_bound`).